

AWS STUDENT BUILDER GROUP

HACKQUEST 2026

Challenge Catalog and Chaos Twists

40 server-owned problem statements across 10 tracks, with one curated Chaos adaptation per challenge.

How to use this catalog

Each team receives one balanced, randomized challenge. Every challenge carries a fixed 100-point reveal award. Before Chaos Mode is revealed, the team leader may make one exact challenge swap for 100 points. Chaos Mode then reveals the challenge-specific adaptation shown in this guide.

10 Tracks

40 Challenges

100 Points Each

40 Chaos Twists

AI & Automation

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Adaptive Campus Services Concierge		100 PTS
REQUIRED OUTCOME	Personalize campus notices, services, and next actions for distinct student needs.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Manual Override Required Add a manual override and record why a recommendation changed.	

2. Document-to-Workflow Automator		100 PTS
REQUIRED OUTCOME	Turn forms or requests into a tracked multi-step workflow.	
SUGGESTED AWS SERVICES	Lambda, Step Functions, DynamoDB	
CHAOS ADAPTATION	Second Input Format Support a second input format, including scanned forms.	

3. Feedback Signal-to-Action Hub		100 PTS
REQUIRED OUTCOME	Convert feedback into evidence-backed issues and actions.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Action Notification Notify the responsible role with the evidence that triggered it.	

4. Human-in-the-Loop Review Assistant		100 PTS
REQUIRED OUTCOME	Help reviewers prioritize work while humans approve final actions.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, Cognito	
CHAOS ADAPTATION	Low Confidence Escalation Escalate low-confidence suggestions for manual review.	

Healthcare & Wellbeing

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Safe Prescription Organizer		100 PTS
REQUIRED OUTCOME	Digitize prescription details and flag interaction concerns for pharmacist review without giving medical advice.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Caregiver Access Add approved caregiver access to medication schedules.	

2. Wellbeing Pattern Explorer		100 PTS
REQUIRED OUTCOME	Explain trends from synthetic or user-provided activity data.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Incomplete Data Handle missing data and communicate uncertainty.	

3. Everyday Care Coordination Hub		100 PTS
REQUIRED OUTCOME	Coordinate tasks, reminders, logs, and updates for care roles.	
SUGGESTED AWS SERVICES	Cognito, DynamoDB, SNS	
CHAOS ADAPTATION	Urgent Task Escalation Escalate an urgent task when its owner does not respond.	

4. Sports Load & Recovery Planner		100 PTS
REQUIRED OUTCOME	Recommend workload adjustments from non-clinical training data.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, EventBridge	
CHAOS ADAPTATION	Consent Controls Add consent and user-controlled data deletion.	

FinTech & Digital Commerce

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Fraud Network Explorer		100 PTS
REQUIRED OUTCOME	Surface suspicious synthetic transaction networks for reviewer action.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Evidence Trail Required Show a plain-language evidence trail for every alert.	

2. Student Budget & Cashflow Coach		100 PTS
REQUIRED OUTCOME	Categorize spending and forecast progress toward a savings target.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Offline Reconciliation Import offline CSV data and reconcile duplicates.	

3. Explainable Payment Risk Guard		100 PTS
REQUIRED OUTCOME	Score synthetic payment risk and explain extra-verification decisions.	
SUGGESTED AWS SERVICES	API Gateway, Lambda, DynamoDB	
CHAOS ADAPTATION	Step-up Verification Add a safe step-up verification flow.	

4. Small Business Invoice Anomaly Monitor		100 PTS
REQUIRED OUTCOME	Detect duplicate, unusual, or rapidly changing invoice patterns.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Merchant Dispute Workflow Add a merchant dispute workflow with preserved history.	

Smart Cities, Mobility & Disaster Response

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Emergency Green-Corridor Planner		100 PTS
REQUIRED OUTCOME	Recommend safer emergency routes from simulated traffic data.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, API Gateway	
CHAOS ADAPTATION	Road Closure Recalculate when a road closure invalidates the route.	

2. Campus Shuttle Demand Planner		100 PTS
REQUIRED OUTCOME	Help students choose less-crowded shuttle options.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, EventBridge	
CHAOS ADAPTATION	Demand Surge React to an event ending early and changed demand.	

3. Disaster Incident Triage Map		100 PTS
REQUIRED OUTCOME	Deduplicate, classify, and prioritize synthetic incident reports.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Communication Outage Queue and later synchronize reports during an outage.	

4. Relief Resource Allocation Board		100 PTS
REQUIRED OUTCOME	Match supplies and volunteers to high-priority requests.	
SUGGESTED AWS SERVICES	SQS, SNS, Lambda	
CHAOS ADAPTATION	Depot Unavailable Reallocate after a key supply depot becomes unavailable.	

Cybersecurity, Trust & Digital Identity

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Cloud Drift Detection Simulator		100 PTS
REQUIRED OUTCOME	Detect simulated baseline violations and recommend remediation.	
SUGGESTED AWS SERVICES	CloudWatch, Lambda, DynamoDB	
CHAOS ADAPTATION	New Security Baseline Apply a new baseline rule to existing resources.	

2. Zero-Trust API Activity Monitor		100 PTS
REQUIRED OUTCOME	Identify abnormal synthetic API usage with evidence.	
SUGGESTED AWS SERVICES	API Gateway, Lambda, DynamoDB	
CHAOS ADAPTATION	Risk Explanations Add per-user risk explanations, not just block or allow.	

3. Privacy Consent & Data Control Center		100 PTS
REQUIRED OUTCOME	Let users review, modify, export, and revoke consent.	
SUGGESTED AWS SERVICES	Cognito, DynamoDB, S3	
CHAOS ADAPTATION	Consent Revocation Process mid-flow consent revocation with an audit record.	

4. Media Authenticity Evidence Desk		100 PTS
REQUIRED OUTCOME	Assess media or claims using sources and uncertainty.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Conflicting Sources Present conflicting evidence and unresolved confidence.	

Education, Skills & Accessibility

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Accessible Lecture Companion		100 PTS
REQUIRED OUTCOME	Deliver captions, notes, and accessible visual descriptions.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Keyboard-first Access Make the core workflow keyboard and screen-reader ready.	

2. Knowledge-Gap Diagnostic		100 PTS
REQUIRED OUTCOME	Map assessments to actionable concept gaps.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Incomplete Assessment Lower confidence when assessment data is incomplete.	

3. Adaptive Practice Studio		100 PTS
REQUIRED OUTCOME	Change practice tasks based on learner performance.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Second Learner Profile Support a second learner with a distinct next path.	

4. Skills-to-Career Navigator		100 PTS
REQUIRED OUTCOME	Map skills to a realistic target-role learning plan.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Target Role Changed Recompute and explain the plan after the target role changes.	

Climate, Energy & Agriculture

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Microgrid Energy Planner		100 PTS
REQUIRED OUTCOME	Recommend stable energy storage actions from simulated data.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, EventBridge	
CHAOS ADAPTATION	Forecast Feed Failure Fall back safely when forecast data fails.	

2. Water Anomaly & Leak Watch		100 PTS
REQUIRED OUTCOME	Detect abnormal water use and recommend actions.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, SNS	
CHAOS ADAPTATION	Major Leak Alert Escalate a major leak to resident and facilities roles.	

3. Food Waste Redistribution Hub		100 PTS
REQUIRED OUTCOME	Match usable food surplus to verified nearby demand.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Donor Cancelled Reassign stock when a donor cancels after matching.	

4. Crop Health & Advisory Assistant		100 PTS
REQUIRED OUTCOME	Give non-binding, localized crop-care guidance.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Weather Risk Changed Update advice after weather-risk conditions change.	

Retail, Supply Chain & Logistics

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Inventory Shrinkage & Demand Forecasting Engine		100 PTS
REQUIRED OUTCOME	Forecast demand and flag stock anomalies.	
SUGGESTED AWS SERVICES	Lambda, S3, DynamoDB	
CHAOS ADAPTATION	Conflicting Stock Count Reconcile conflicting stock counts with confidence.	

2. Cold-Chain Compliance Tracker		100 PTS
REQUIRED OUTCOME	Track synthetic custody and temperature compliance.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Telemetry Loss Quarantine and review a batch after telemetry loss.	

3. Low-Carbon Fleet Dispatch Planner		100 PTS
REQUIRED OUTCOME	Balance delivery capacity, delay, and emissions.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, API Gateway	
CHAOS ADAPTATION	Vehicle Unavailable Re-plan after one vehicle becomes unavailable.	

4. Explainable Size & Fit Assistant		100 PTS
REQUIRED OUTCOME	Recommend size and products using purchase and return history.	
SUGGESTED AWS SERVICES	API Gateway, Lambda, DynamoDB	
CHAOS ADAPTATION	Return-rate Spike Adapt after a return-rate spike and explain why.	

Workforce, Productivity & Digital Life

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Meeting-to-Action Compiler		100 PTS
REQUIRED OUTCOME	Extract owners, deadlines, and decisions from transcripts.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Missed Action Escalate and reassign a missed action.	

2. Workload Balance Monitor		100 PTS
REQUIRED OUTCOME	Identify unsustainable work patterns without clinical claims.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, EventBridge	
CHAOS ADAPTATION	Missing Self-report Handle stopped self-reporting with visible uncertainty.	

3. Team Knowledge Finder		100 PTS
REQUIRED OUTCOME	Retrieve reliable project information with source context.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Conflicting Documents Resolve conflicting documents using freshness and review.	

4. Project Coordination Pulse		100 PTS
REQUIRED OUTCOME	Show dependencies, blockers, and shared team status.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, SNS	
CHAOS ADAPTATION	Contributor Unavailable Rebalance work when a contributor becomes unavailable.	

Civic Tech & Open Innovation

Each challenge includes an original required outcome, suggested AWS services, and a curated adaptation that is revealed only when Chaos Mode begins.

1. Public Grievance Triage Engine		100 PTS
REQUIRED OUTCOME	Route civic complaints and track resolution against an SLA.	
SUGGESTED AWS SERVICES	S3, Lambda, DynamoDB	
CHAOS ADAPTATION	Duplicate Report Surge Merge a surge of duplicate reports while retaining evidence.	

2. Participatory Budgeting Tool		100 PTS
REQUIRED OUTCOME	Rank proposals and simulate transparent budget allocation.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Budget Cut Recompute after the available budget falls by 20%.	

3. Low-Connectivity Public Service Portal		100 PTS
REQUIRED OUTCOME	Support a civic process on poor connectivity and low-end devices.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, S3	
CHAOS ADAPTATION	Offline Sync Conflict Queue offline submissions and resolve a sync conflict.	

4. Inclusive Campus Service Redesign		100 PTS
REQUIRED OUTCOME	Improve a defined service for an underserved user group.	
SUGGESTED AWS SERVICES	Lambda, DynamoDB, API Gateway	
CHAOS ADAPTATION	Second User Persona Support a second persona with different language or accessibility needs.	